

DNA Replication in Prokaryotes and Eukaryotes

A Complete Molecular Biology Teaching and Learning Module

Laboratory Workbook

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DNA Replication — Laboratory Workbook

Course: Molecular Biology of DNA Replication Student Name:
Date: Lab Partner:
Section:

Course Information

Instructor: Email: Lab Room:
Lab Time:

Learning Outcomes

Upon completing all laboratory exercises, students will be able to:

1. Model DNA replication structures and processes
2. Quantify and assess DNA quality
3. Perform agarose gel electrophoresis
4. Conduct and analyze PCR experiments
5. Interpret replication-related data
6. Apply computational approaches to replication analysis

Laboratory Rules

1. **Arrive on time** — pre-lab preparation is required
2. **Wear PPE** — lab coat, gloves, safety glasses at all times in wet lab
3. **No food or drink** in the laboratory
4. **Clean up** your workspace after each experiment
5. **Report all accidents** immediately to the instructor
6. **Follow SOPs** — Standard Operating Procedures must be followed for all procedures
7. **Dispose of waste** properly — follow institutional waste disposal guidelines
8. **No unauthorized experiments** — only perform approved procedures
9. **Record everything** — keep detailed notes in this workbook
10. **Academic integrity** — your work must be your own

Safety Summary

Chemical Hazards

Chemical	Hazard	PPE Required
Ethidium bromide	Mutagen	Double gloves, lab coat
Agarose (hot)	Burn hazard	Heat-resistant gloves
TAE buffer	Low hazard	Gloves
Loading dye	Low hazard	Gloves

Biological Hazards

- DNA samples: Follow institutional biosafety guidelines
- PCR products: Handle with care to prevent contamination

Equipment Hazards

- Electrophoresis: Electrical hazard — never touch running equipment
- Microwave: Burn hazard — handle hot flasks with care
- UV transilluminator: Eye hazard — use protective goggles

Emergency Procedures

- **Spill:** Notify instructor, clean up per SOP
- **EtBr exposure:** Wash thoroughly, notify instructor
- **Electrical shock:** Disconnect power, seek medical attention
- **Eye exposure:** Flush with water for 15 minutes, seek medical attention

EXPERIMENT 1: DNA Structure and Replication Modeling

Date: _____ **Time Started:** _____ **Time Ended:** _____

Pre-Lab Questions

1. What are the four nucleotide bases in DNA?
2. Which bases pair together, and how many hydrogen bonds form between each pair?
3. What does "antiparallel" mean in the context of DNA structure?
4. What is the difference between the 5' and 3' ends of a DNA strand?
5. Why is the direction of DNA synthesis important?

Materials Checklist

- ☐ Colored paper strips (4 colors)
- ☐ Scissors
- ☐ Tape/glue
- ☐ Markers
- ☐ Printed diagram

Procedure Notes

(Record any observations or modifications during the procedure)

Part A: DNA Structure Recognition

My 20-bp DNA sequence:

Top strand (5'→3'): _____

Bottom strand (5'→3'): _____

Number of A-T pairs: _____ Number of G-C pairs: _____

Total hydrogen bonds: _____

Part B: Replication Fork Construction

Direction of fork movement: _____

Template for leading strand: _____

Template for lagging strand: _____

Part C: Strand Synthesis

Leading Strand

Feature	My Model	Expected
Number of primers		1
Continuity		Continuous
Direction (toward/away from fork)		Toward fork

Lagging Strand

Feature	My Model	Expected
Number of Okazaki fragments		3-4
Size of fragments		Equal
Direction of each fragment		5'→3'

Feature	My Model	Expected
Overall direction		Away from fork

Reflection Questions

1. Which was harder to model — leading or lagging strand? Why?
2. What would happen if a mutation prevented Okazaki fragment joining?
3. How does the antiparallel nature of DNA necessitate the lagging strand?

EXPERIMENT 2: DNA Quantification and Quality Assessment

Date: _____ Time Started: _____ Time Ended: _____

Pre-Lab Questions

1. At what wavelength does DNA absorb maximally?
2. What does the A₂₆₀/A₂₈₀ ratio tell you about DNA purity?
3. What is the Beer-Lambert law?
4. Why is TE buffer used instead of water for DNA dilution?

Procedure Notes

Spectrophotometric Results

Part A: Standard Curve (if applicable)

Standard	Concentration (ng/μL)	A ₂₆₀
Blank	0	
Std 1		
Std 2		
Std 3		

Part B: Sample Measurements

Sample	A ₂₆₀	A ₂₈₀	A ₂₃₀	A ₂₆₀ /A ₂₈₀	A ₂₆₀ /A ₂₃₀	Conc (ng/μL)	Assessment
1							
2							
3							
4							

Sample	A260	A280	A230	A260/A280	A260/A230	Conc (ng/ μ L)	Assessment
5							

Calculations

Example calculation for Sample 1:

Concentration = A260 $\times$ DF $\times$ 50 μ g/mL

= _____ $\times$ _____ $\times$ 50 = _____ μ g/mL = _____ ng/ μ L

Interpretation

1. Which sample had the best quality? _____
2. Which sample had the worst quality? _____
3. What contamination might explain poor A260/A280 ratio?
4. What would you do to improve the quality of a contaminated sample?

EXPERIMENT 3: Agarose Gel Electrophoresis

Date: _____ Time Started: _____ Time Ended: _____

Pre-Lab Questions

1. Why does DNA migrate toward the anode (+) during electrophoresis?
2. How does agarose concentration affect DNA separation?
3. What is the purpose of loading dye?
4. At what concentration of agarose would you separate 200 bp fragments?

Gel Preparation

Agarose concentration: _____ % Volume: _____ mL

Buffer: _____ Voltage: _____ V

Time run: _____ minutes

Migration Distances

Molecular Weight Marker

Band Size (bp)	Distance Migrated (cm)	$\log_{10}(\text{size})$
500		
1,000		
2,000		

Band Size (bp)	Distance Migrated (cm)	log ₁₀ (size)
3,000		
4,000		
5,000		
6,000		
8,000		
10,000		

Sample Results

Lane	Sample	Migration Distance (cm)	Estimated Size	Band Quality
1	Ladder	—	—	
2				
3				
4				
5				
6				

Gel Diagram

(Draw your gel here, including marker bands and sample bands)

Lane: 12345678

M S1 S2 S3 S4 S5 – M

Interpretation

1. Were the results as expected? Explain any discrepancies.
2. What might a smear indicate about DNA quality?
3. How would you improve resolution of small fragments?

EXPERIMENT 4: PCR and DNA Amplification

Date: _____ Time Started: _____ Time Ended: _____

Pre-Lab Questions

1. How is PCR different from cellular DNA replication?
2. What temperature is used for each PCR step, and why?
3. Why are two primers needed for PCR?
4. What is the theoretical yield after 30 cycles?
5. What is the purpose of the negative control?

PCR Setup

Master Mix Preparation

Component	Volume per reaction (μL)	Volume for 3 reactions (μL)
10× buffer	5.0	
dNTP mix	1.0	
Forward primer	1.0	
Reverse primer	1.0	
Taq polymerase	0.25	
Water	40.75	
Total	49.0	

PCR Program

Step	Temperature (°C)	Time	Cycles
Initial denaturation			1
Denaturation			
Annealing			
Extension			
Final extension			1

Results

Gel Analysis

Lane	Sample	Expected Size	Observed Size	Band Intensity	Interpretation
M	Ladder	—	—	—	
1	Positive				
2	Negative				

Lane	Sample	Expected Size	Observed Size	Band Intensity	Interpretation
3					

Gel Diagram

(Draw your PCR gel here)

Interpretation

1. Was the PCR successful? Evidence?
2. What does the negative control result tell you?
3. If you got no band in the positive sample, list 3 possible causes:
 - a. _____
 - b. _____
 - c. _____
4. Calculate the theoretical yield if you started with 10 template molecules and ran 30 cycles:

EXPERIMENT 5: Replication Data Analysis

Date: _____ Time Started: _____ Time Ended: _____

Exercise 1: DNA Fiber Analysis

Fiber Track Measurements

Fiber #	IdU track (μm)	CldU track (μm)	Pattern
1			
2			
3			
4			
5			

Fork Speed Calculation

Average track length: _____ μm

Conversion: 1 $\mu\text{m} \approx 2 \text{ kb} \rightarrow$ _____ kb

Speed = _____ kb / 20 min = _____ kb/min = _____ nt/s

Questions

1. What pattern did you observe most frequently? _____
2. What do asymmetric tracks tell you? _____

3. How would you interpret shorter CldU tracks in drug-treated cells?

Exercise 2: Okazaki Fragment Analysis

Sample	Average Fragment Size	Assessment

Exercise 3: GC-Skew Analysis

Position (kb)	GC-skew	Interpretation

Origin predicted at position: _____ kb

Exercise 4: Replication Timing

Gene	Expression	Timing	Correlation

EXPERIMENT 6: Capstone Exercise

Date: _____ Time Started: _____ Time Ended: _____

Part A: E. coli Replication Mutant

Hypothesis:

Evidence supporting hypothesis:

Experimental design to test hypothesis:

Predicted results:

Part B: Cancer Patient Scenario

Diagnosis:

Molecular basis:

Treatment recommendation:

Genetic counseling recommendation:

Part C: Telomere Analysis

Explanation:

Clinical significance:

Part D: Drug Mechanism

Target prediction:

Evidence:

Therapeutic strategy:

Post-Course Reflection

What I learned:

- 1.
- 2.
- 3.
- 4.
- 5.

Skills I developed:

- 1.
- 2.
- 3.

Areas for improvement:

- 1.
- 2.

Instructor Sign-off

Experiment	Grade	Instructor Initials	Date
Lab 1			
Lab 2			
Lab 3			
Lab 4			
Lab 5			
Lab 6			

Experiment	Grade	Instructor Initials	Date
Overall			

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Lab 01: DNA Structure and Replication Modeling

Experiment Title

DNA Structure Recognition and Replication Fork Modeling

Learning Objectives

1. Identify key structural features of the DNA double helix relevant to replication
2. Construct and interpret a model of the replication fork
3. Demonstrate understanding of leading and lagging strand synthesis
4. Identify the roles of major replication proteins

Background Theory

DNA replication requires understanding of:

- Antiparallel strand orientation (5'→3' and 3'→5')
- Complementary base pairing (A-T, G-C)
- Semiconservative replication
- Replication fork structure (Y-shaped)
- Okazaki fragments on the lagging strand

Materials

For Modeling Exercise

Material	Quantity	Notes
Colored paper strips (4 colors)	8 per group	Represent nucleotides (A, T, G, C)
Scissors	1 per group	
Tape or glue	As needed	
Markers/pens	As needed	
Printed replication fork diagram	1 per group	Reference

Equipment

- None required (paper-based modeling)

Safety Considerations

- Standard classroom safety only
- Handle scissors carefully

Experimental Principle

The replication fork is a Y-shaped structure where:

- Helicase unwinds the double helix
- Leading strand is synthesized continuously toward the fork
- Lagging strand is synthesized discontinuously away from the fork as Okazaki fragments
- Each Okazaki fragment requires its own RNA primer

Step-by-Step Procedure

Part A: DNA Structure Recognition (15 minutes)

1. Using colored paper strips, create a 20-base-pair double-stranded DNA molecule
 - Use 4 colors for A, T, G, C
 - Ensure complementary base pairing
 - Write 5' and 3' labels on each strand
 - Ensure strands are antiparallel
2. Answer the following:
 - What is the sequence of the top strand? 5'→3'
 - What is the sequence of the bottom strand? 5'→3'
 - How many hydrogen bonds between A-T? Between G-C?
 - Which end would DNA polymerase read from?

Part B: Replication Fork Construction (20 minutes)

3. Cut your DNA model at the middle to represent helicase unwinding
4. Separate the strands to create a Y-shaped fork
5. Label the following on your model:
 - 5' and 3' ends of each strand
 - Template strand for leading strand
 - Template strand for lagging strand
 - Direction of fork movement

Part C: Leading and Lagging Strand Synthesis (25 minutes)

6. Using new colored strips, synthesize the leading strand:
 - Identify which template strand to use
 - Build the new strand 5'→3' toward the fork

- Only one primer needed (use a different color for RNA primer)

7. Synthesize Okazaki fragments on the lagging strand:

- Use a different color for each fragment
- Each fragment should have an RNA primer (different color) + DNA
- Build fragments 5'→3' but away from the fork
- Make 3-4 fragments

8. Label:

- Leading strand (continuous)
- Lagging strand (discontinuous)
- Okazaki fragments
- RNA primers
- Direction of synthesis on each strand

Expected Observations

Observation	Expected Result
Leading strand	Single continuous strand, one primer at start
Lagging strand	Multiple fragments, each with primer
Fragment orientation	Each Okazaki fragment 5'→3'
Overall lagging direction	Opposite to fork movement
Complementarity	New strands complementary to templates

Results Table

Feature	Leading Strand	Lagging Strand
Synthesis direction	5'→3' (toward fork)	5'→3' (away from fork)
Continuity	Continuous	Discontinuous
Number of primers	1	Multiple (one per fragment)
Okazaki fragments	None	Yes (3-4 in model)
Template strand	3'→5' (toward fork)	5'→3' (toward fork)

Calculations

Fork Speed Calculation

If the replication fork moves at 1,000 nt/second in *E. coli*:

Question: How long would it take to replicate a 2,000 bp Okazaki fragment?

Answer:

- Speed = 1,000 nt/second
- Okazaki fragment = 2,000 nt
- Time = 2,000 / 1,000 = **2 seconds**

Question: If *E. coli* genome is 4.6×10^6 bp and replication is bidirectional from one origin, how many nucleotides does each fork copy?

Answer:

- Each fork copies half = 2.3×10^6 nt
- Time = 2.3×10^6 / 1,000 = 2,300 seconds $\approx$ **38 minutes**

Interpretation

1. Why can't both strands be synthesized continuously?
2. What would happen if Okazaki fragments were not joined?
3. How does the antiparallel nature of DNA necessitate the lagging strand mechanism?

Troubleshooting

Problem	Possible Cause	Solution
Fragments not oriented correctly	Confusion about 5'→3' direction	Re-check polarity of template strand
Leading strand has multiple primers	Confused leading/lagging	Leading strand only needs one primer
Fragments too large	Not following template	Ensure complementary base pairing

Post-Lab Questions

1. In your model, what protein would remove the RNA primers? (See Module 03)
2. If a mutation prevented Okazaki fragment ligation, what would be the consequence?
3. Why is it important that DNA polymerase only synthesizes in the 5'→3' direction?
4. Draw a diagram of your completed replication fork model.

Viva Questions

1. What is the direction of DNA synthesis? Why?
2. Explain why the lagging strand is called "lagging."
3. What would happen if helicase worked in only one direction?
4. How many primers are needed per replication fork?
5. What is the role of the sliding clamp?

Expected Answers

1. **5'→3' direction** — DNA polymerase can only add nucleotides to the free 3'-OH group.
2. **Lagging strand** is synthesized in the direction opposite to fork movement, making it "lag behind" the leading strand in overall synthesis.

3. **If helicase worked unidirectionally** — only one fork would form, doubling replication time.
4. **Two primers per fork** — one for leading strand, one initially for lagging strand (plus new primers for each Okazaki fragment).
5. **Sliding clamp** increases polymerase processivity by tethering it to the DNA template.

Instructor Notes

- Common student error: confusing template strand polarity with synthesis direction
- Allow time for students to physically build the model before abstracting
- Use this lab to set up the conceptual foundation for all subsequent labs
- This is a modeling/discussion exercise, not a wet-lab experiment
- Connect to Module 01 and Module 02 lecture content

Related Theory

- [Module 01: Introduction](#)
- [Module 02: Core Concepts](#)
- [Cheat Sheet](#)

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Lab 02: DNA Quantification and Quality Assessment

Experiment Title

DNA Quantification Using Spectrophotometry and Fluorometry

Learning Objectives

1. Quantify DNA concentration using UV spectrophotometry
2. Assess DNA purity using A₂₆₀/A₂₈₀ and A₂₆₀/A₂₃₀ ratios
3. Understand the principles of Beer-Lambert law
4. Perform quality assessment of extracted DNA
5. Compare spectrophotometric and fluorometric quantification methods

Background Theory

Beer-Lambert Law

$A = \epsilon lc$ Where:

- A = Absorbance
- ϵ = Molar extinction coefficient (for dsDNA: 0.020 cm⁻¹ at 260 nm for 50 µg/mL)
- l = Path length (1 cm)
- c = Concentration

DNA Absorption

- DNA absorbs UV light maximally at **260 nm** (due to aromatic bases)
- Proteins absorb at **280 nm** (due to aromatic amino acids)
- Contaminants (phenol, guanidine) absorb at **230 nm**

Purity Ratios

Ratio	Acceptable Range	Interpretation
A ₂₆₀ /A ₂₈₀	1.8–2.0	Pure DNA (~1.8); RNA contamination > 1.8
A ₂₆₀ /A ₂₃₀	>2.0	Pure DNA; <2.0 indicates contamination

Materials

Material	Quantity	Notes
Genomic DNA samples (various qualities)	3-5	Pre-extracted
TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	100 mL	Diluent
UV spectrophotometer (NanoDrop or equivalent)	1	
Fluorometer (Qubit or equivalent)	1	If available
Microvolume cuvettes	As needed	
Clean pipette tips	As needed	
DEPC-treated water	50 mL	

Equipment

- UV-Vis spectrophotometer (NanoDrop preferred)
- Fluorometer (if available)
- Micropipettes (P20, P200, P1000)
- Vortex mixer
- Microcentrifuge

Safety Considerations

⚠️ **UV light hazard:** Never look directly at UV light source ⚠️ **Chemical hazard:** Wear gloves when handling DNA samples ⚠️ **Biological hazard:** Follow institutional biosafety guidelines ⚠️ Always use appropriate PPE (lab coat, gloves, safety glasses)

Experimental Principle

DNA absorbs UV light at 260 nm due to the conjugated ring systems of purine and pyrimidine bases. The absorbance is proportional to DNA concentration (Beer-Lambert law). Contaminating proteins, RNA, and chemicals absorb at different wavelengths, allowing purity assessment.

Step-by-Step Procedure

Part A: Spectrophotometric Quantification (30 minutes)

1. Blank the spectrophotometer

- Use 1 μ L of TE buffer (same as diluent)
- Run blank measurement

2. Measure sample 1 (high-quality genomic DNA)

- Load 1 μ L onto NanoDrop
- Record A₂₆₀, A₂₈₀, A₂₃₀

- Calculate concentration: $A_{260} \times \text{dilution factor} \times 50 \mu\text{g/mL}$

3. Measure samples 2-5

- Repeat for all samples
- Record all values in results table

4. Calculate purity ratios

- A_{260}/A_{280}
- A_{260}/A_{230}

Part B: Fluorometric Quantification (if available) (20 minutes)

5. Prepare Qubit working solution

- Dilute Qubit reagent 1:200 in Qubit buffer
- Add 199 μL buffer + 1 μL reagent per tube

6. Prepare standards

- Standard 1: 0 ng/ μL (blank)
- Standard 2: 10 ng/ μL (high standard)

7. Prepare samples

- Dilute 1-10 μL sample in 200 μL working solution
- Include a "sample blank" (working solution only)

8. Read standards and samples

- Follow instrument protocol
- Record concentrations

Expected Observations

Sample	A260	A280	A230	A260/A280	A260/A230	Quality
Sample 1 (high quality)	~1.0	~0.55	~0.40	~1.85	~2.5	Excellent
Sample 2 (RNA contam.)	~1.2	~0.55	~0.50	~2.1	~2.4	RNA contamination
Sample 3 (Protein contam.)	~0.8	~0.50	~0.50	~1.6	~1.6	Protein contamination
Sample 4 (Low concentration)	~0.1	~0.06	~0.05	~1.7	~2.0	Low but pure
Sample 5 (Multiple contam.)	~0.5	~0.35	~0.30	~1.4	~1.7	Multiple contaminants

Results Table

Sample	Conc. (ng/ μL)	A260/A280	A260/A230	Fluorometer Conc.	Assessment
1					
2					
3					

Sample	Conc. (ng/ μ L)	A260/A280	A260/A230	Fluorometer Conc.	Assessment
4					
5					

Calculations

Concentration Calculation

For a NanoDrop reading:

- Dilution factor (DF): volume of sample + volume of diluent
- Concentration = $A_{260} \times DF \times 50 \text{ } \mu\text{g/mL}$

Example:

- $A_{260} = 0.5$
- $DF = 1$ (undiluted)
- Concentration = $0.5 \times 1 \times 50 = 25 \text{ } \mu\text{g/mL} = 25,000 \text{ ng}/\mu\text{L}$

Total Yield

Total DNA = Concentration $\times$ Total volume

Example:

- Concentration = $25 \text{ } \mu\text{g/mL}$
- Total volume = $200 \text{ } \mu\text{L} = 0.2 \text{ mL}$
- Total DNA = $25 \times 0.2 = 5.0 \text{ } \mu\text{g}$

Interpretation

1. Why does RNA contamination increase A260/A280 above 1.8?
2. Why does protein contamination decrease A260/A280 below 1.8?
3. How would you use this DNA for PCR? What concentration is optimal?
4. When would you prefer fluorometric over spectrophotometric quantification?

Troubleshooting

Problem	Possible Cause	Solution
A260 reading too high (>3.0)	Too concentrated	Dilute sample and re-measure
A260 reading too low (<0.05)	Too dilute	Use fluorometer or concentrate sample
$A_{260}/A_{280} < 1.6$	Protein contamination	Re-extract with proteinase K
$A_{260}/A_{280} > 2.0$	RNA contamination	Treat with RNase A
$A_{260}/A_{230} < 1.5$	Salt/phenol contamination	ethanol precipitation or clean-up column
Negative A260	Bubbles or dirty cuvette	Clean and re-measure

Post-Lab Questions

1. A DNA sample has $A_{260} = 0.8$, $A_{280} = 0.5$, $A_{230} = 0.4$. What is the concentration? Is it pure?
2. Why is TE buffer used instead of water for DNA dilution?
3. When would you use fluorometric quantification instead of spectrophotometric?
4. How does the Beer-Lambert law relate to DNA quantification?
5. A sample has $A_{260}/A_{280} = 2.3$. What contamination might this indicate?

Viva Questions

1. What is the maximum absorbance wavelength for DNA? For proteins?
2. Why do you blank the instrument before measuring samples?
3. What is the path length of a NanoDrop?
4. How does the A_{260}/A_{280} ratio change with protein contamination?
5. Name two contaminants that absorb at 230 nm.

Expected Answers

1. **Concentration:** $0.8 \times 1 \times 50 = 40 \mu\text{g/mL}$. **Purity:** $A_{260}/A_{280} = 1.6$ (low, protein contamination); $A_{260}/A_{230} = 2.0$ (acceptable). Overall: impure DNA with protein contamination.
2. **TE buffer** chelates divalent cations (Mg^{2+}) that activate DNases, protecting DNA from degradation.
3. **Fluorometric** is preferred for low-concentration samples, crude preparations, or when contaminants interfere with UV absorbance.
4. **Beer-Lambert law** states that absorbance is directly proportional to concentration (at constant path length and extinction coefficient).
5. **Contaminants at 230 nm:** phenol, guanidine thiocyanate, EDTA, carbohydrates, salt.

Instructor Notes

- This is a wet-lab exercise requiring a spectrophotometer
- Ensure NanoDrop is cleaned before use
- Prepare DNA samples of varying quality in advance
- Discuss real-world scenarios where DNA quality matters (PCR, sequencing, cloning)
- If fluorometer is not available, focus on spectrophotometric methods

Related Theory

- [Module 09: Experimental Methods](#)
- [Lab 03: Gel Electrophoresis](#)

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Lab 03: Agarose Gel Electrophoresis

Experiment Title

Agarose Gel Electrophoresis for DNA Fragment Analysis

Learning Objectives

- 1. Prepare and cast agarose gels
- 2. Load and run DNA samples on an agarose gel
- 3. Visualize DNA fragments using ethidium bromide or SYBR Safe
- 4. Estimate DNA fragment sizes using a molecular weight marker
- 5. Interpret gel electrophoresis results

Background Theory

Agarose gel electrophoresis separates DNA fragments by size. DNA is negatively charged (phosphate backbone) and migrates toward the positive electrode (anode). Smaller fragments migrate faster through the agarose matrix than larger fragments.

Key Principles

- **Migration:** DNA moves toward + electrode (anode)
- **Separation by size:** Smaller = faster, Larger = slower
- **Resolution:** Depends on agarose concentration
- **Visualization:** Intercalating dyes (EtBr, SYBR Safe) fluoresce under UV/blue light

Agarose Concentration Guide

Agarose %	Optimal Size Range	Resolution
0.7%	0.8–10 kb	Large fragments
1.0%	0.5–7 kb	General purpose
1.5%	0.2–4 kb	Medium fragments
2.0%	0.1–2 kb	Small fragments

Materials

Material	Quantity	Notes
Agarose powder	1.5 g	For 1.0% gel in 150 mL

Material	Quantity	Notes
1× TAE buffer	200 mL	Running and gel buffer
Ethidium bromide (10 mg/mL)	10 µL	Intercalating dye (handle with care)
OR SYBR Safe	10 µL	Safer alternative
6× DNA loading dye	10 µL	Contains glycerol + tracking dyes
DNA molecular weight marker (1 kb ladder)	5 µL	Size standard
DNA samples	5-10 µL each	Various sizes/qualities
Distilled water	As needed	

Equipment

- Microwave or agarose gel casting system
- Gel casting tray and comb
- Horizontal gel electrophoresis chamber
- DC power supply (5-10 V/cm)
- UV transilluminator or blue light illuminator
- Gel documentation system
- Micropipettes (P20, P200)
- UV-protective goggles

Safety Considerations

⚠ **Ethidium bromide is a mutagen** — handle with gloves at all times ⚠ **Hot agarose** — allow to cool before pouring ⚠ **UV light** — use transilluminator with protective shield; wear UV goggles ⚠ **Electrical hazard** — ensure electrophoresis chamber is properly connected ⚠ Follow institutional safety guidelines for chemical and biological hazards

Experimental Principle

DNA migration through agarose is governed by:

- **Electric field strength:** Voltage drives migration
- **Agarose concentration:** Higher % = smaller pores = slower migration of large DNA
- **DNA size:** Log-linear relationship between size and migration distance
- **DNA conformation:** Supercoiled > linear > nicked circular (for same size)

Step-by-Step Procedure

Part A: Gel Preparation (15 minutes + cooling)

1. Weigh 1.5 g agarose powder
2. Add to 150 mL of 1× TAE buffer in a flask

3. Microwave to dissolve agarose (2-3 minutes, with swirling)
4. Allow to cool to $\sim 55^{\circ}\text{C}$ (warm but comfortable to hold flask)
5. Add 10 μL ethidium bromide (or SYBR Safe) per 150 mL gel
6. Pour into casting tray with comb in place
7. Allow to solidify (20-30 minutes at room temperature)
8. Remove comb carefully
9. Place gel in electrophoresis chamber with $1\times$ TAE running buffer

Part B: Sample Preparation and Loading (10 minutes)

10. Mix each DNA sample with loading dye:
 - 5 μL DNA + 1 μL $6\times$ loading dye
11. Load samples into wells:
 - Lane 1: Molecular weight marker (5 μL)
 - Lane 2: Sample 1 (6 μL)
 - Lane 3: Sample 2 (6 μL)
 - Lane 4: Sample 3 (6 μL)
 - Lane 5: Sample 4 (6 μL)
 - Lane 6: Sample 5 (6 μL)
 - Lane 7: Empty (control)
 - Lane 8: Molecular weight marker
12. Close gel box, connect electrodes (black = cathode, red = anode)

Part C: Electrophoresis (30-45 minutes)

13. Run at 80-100 V (5-8 V/cm) until blue dye front is $\sim 2/3$ down the gel
14. Turn off power supply
15. Remove gel carefully

Part D: Visualization (10 minutes)

16. Place gel on UV transilluminator
17. Visualize bands using UV light (302 nm) or blue light (470 nm)
18. Photograph gel with gel documentation system
19. Measure migration distances of marker bands and samples

Expected Observations

Lane	Sample	Expected Pattern
1, 8	1 kb ladder	Multiple bands at known sizes (0.5, 1, 2, 3, 4, 5, 6, 8, 10 kb)
2	Genomic DNA	High MW band near well (>20 kb)
3	PCR product (single)	Single sharp band at expected size

Lane	Sample	Expected Pattern
4	Plasmid DNA	Multiple bands (supercoiled, nicked, linear)
5	Digested plasmid	Bands at expected restriction fragment sizes
6	Degraded DNA	Smear from high to low MW

Results Table

Marker Band (bp)	Migration Distance (cm)
500	
1,000	
2,000	
3,000	
4,000	
5,000	
6,000	
8,000	
10,000	

Sample	Migration Distance	Estimated Size	Quality Assessment
1			
2			
3			
4			
5			

Calculations

Standard Curve Construction

1. Plot **$\log_{10}(\text{size in bp})$** vs. **migration distance (cm)** for marker bands
2. Draw best-fit line (should be linear for most of the range)
3. Use line equation to determine size of unknown bands

Example:

- Unknown band migration: 3.5 cm
- From standard curve: $\log_{10}(\text{size}) = 3.2$
- Size = $10^{3.2} \approx 1,585$ bp

Gel Percentage Effect

- Higher % agarose → better resolution of small fragments
- Lower % agarose → better resolution of large fragments
- Always use appropriate % for your expected fragment sizes

Interpretation

1. What does a smear indicate about DNA quality?
2. Why might a plasmid show multiple bands?
3. How does agarose concentration affect resolution?
4. What would happen if you ran the gel at too high a voltage?

Troubleshooting

Problem	Possible Cause	Solution
Bands are fuzzy	Voltage too high; gel too concentrated	Lower voltage; use lower % gel
No bands visible	Insufficient EtBr; too little DNA; gel too thick	Add more EtBr; increase DNA; use thinner gel
Bands distorted	Uneven gel; wells damaged	Pour gel evenly; load carefully
DNA stuck in wells	High MW DNA; incomplete denaturation	Use pulsed-field for very large DNA
Bands run crooked	Uneven buffer; gel not level	Ensure even buffer; level gel during casting
Bloomed bands	Too much DNA loaded	Load less DNA

Post-Lab Questions

1. A DNA sample shows a band at 3.0 kb. What is the estimated size using the standard curve?
2. Why is ethidium bromide added to the gel rather than the running buffer?
3. How would you separate DNA fragments of 500 bp and 600 bp with good resolution?
4. A plasmid preparation shows bands at 3 kb, 6 kb, and 9 kb. What might these represent?
5. What is the effect of increasing voltage on band resolution?

Viva Questions

1. Why does DNA migrate toward the anode?
2. What is the purpose of loading dye?
3. How does agarose gel differ from polyacrylamide gel?
4. What is the resolution limit of agarose gel electrophoresis?
5. Name two alternatives to ethidium bromide for DNA visualization.

Expected Answers

1. **3.0 kb** — directly read from the standard curve.

2. **EtBr in gel** allows continuous staining during electrophoresis and eliminates extra staining/destaining steps.
3. **Use 2.0% agarose** — higher percentage provides better resolution for small fragments; run at lower voltage.
4. **3 kb** = supercoiled; **6 kb** = nicked/relaxed circular; **9 kb** = multimer or concatemer.
5. **Increasing voltage** increases migration speed but decreases resolution (bands become fuzzier).

Instructor Notes

- This is a wet-lab exercise requiring electrophoresis equipment
- Ensure students understand gel safety (EtBr handling)
- Prepare samples in advance: genomic DNA, PCR product, plasmid, digested plasmid, degraded DNA
- Discuss common lab scenarios: checking PCR products, verifying cloning
- Connect to replication: Okazaki fragments could be visualized on denaturing gels

Related Theory

- [Module 09: Experimental Methods](#)
- [Module 10: Data Analysis](#)

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Lab 04: PCR and DNA Amplification

Experiment Title

Polymerase Chain Reaction (PCR) as an Experimental Model of DNA Synthesis

Important Distinction: PCR ≠ Cellular DNA Replication

Feature	Cellular Replication	PCR
Unwinding	Helicase enzyme	Heat denaturation (95°C)
Priming	RNA primers (primase)	DNA primers (synthetic)
Polymerase	Pol III/Pol ε/δ (37°C)	Taq/Pfu (optimal 72°C)
Processivity	>50,000 nt (with clamp)	~1,000-5,000 nt (no clamp)
Proofreading	Yes (3'→5' exonuclease)	Taq: No; Pfu: Yes
Okazaki fragments	Yes (lagging strand)	No (linear amplification)
Sliding clamp	Yes (β/PCNA)	No
Regulation	Highly regulated	None (in vitro)
Speed	~50-1000 nt/s	~1,000 nt/min
Starting material	Whole genome	Specific target sequence
Cycles	Once per cell cycle	25-40 cycles

Key point: *PCR conceptually mimics DNA replication (uses primers, dNTPs, polymerase, template) but uses different strategies for unwinding and lacks the complexity of the cellular replisome.*

Learning Objectives

1. Understand the principles of PCR
2. Design and perform a PCR experiment
3. Analyze PCR products by agarose gel electrophoresis
4. Compare PCR with cellular DNA replication
5. Understand applications of PCR in molecular biology

Background Theory

PCR Principle

PCR amplifies a specific DNA sequence through repeated cycles of:

1. **Denaturation** (94-98°C): Separates double-stranded DNA
2. **Annealing** (50-65°C): Primers bind to complementary sequences
3. **Extension** (72°C): Taq polymerase synthesizes new DNA

Exponential Amplification

After n cycles, product = 2^n copies (theoretically)

- 30 cycles → $\sim 10^9$ copies (1 billion)
- The target sequence is amplified exponentially while flanking sequences are amplified linearly

Components

Component	Role	Typical Amount
Template DNA	Source DNA containing target	1-100 ng
Forward primer	Binds one strand, provides 3'-OH	0.2-1 μ M
Reverse primer	Binds other strand, provides 3'-OH	0.2-1 μ M
dNTPs (dATP, dTTP, dGTP, dCTP)	Building blocks	200 μ M each
DNA polymerase (Taq)	Synthesizes DNA	1-2.5 U
MgCl ₂	Cofactor for polymerase	1.5-2.5 mM
Buffer	Provides optimal pH and salts	1×
Nuclease-free water	Volume to 50 μ L	As needed

Materials

Material	Quantity per reaction	Notes
Template DNA (λ phage or human)	1 μ L (10 ng)	Target: specific region
Forward primer (20 μ M)	1 μ L	Specific to target
Reverse primer (20 μ M)	1 μ L	Specific to target
10× PCR buffer	5 μ L	With MgCl ₂
dNTP mix (10 mM each)	1 μ L	
Taq DNA polymerase (5 U/ μ L)	0.25 μ L	
Nuclease-free water	40.75 μ L	

Equipment

- Thermal cycler (PCR machine)
- Micropipettes (P10, P20, P200)
- PCR tubes or 96-well plate
- Agarose gel electrophoresis system
- UV transilluminator

Safety Considerations

⚠ **Ethidium bromide** — mutagen (use gloves, handle carefully) ⚠ **UV light** — use protective goggles ⚠ **Hot surfaces** — thermal cycler reaches 98°C ⚠ **Biological hazard** — follow institutional guidelines for DNA handling ⚠ Wear gloves and lab coat throughout

Experimental Principle

PCR uses thermocycling to achieve exponential amplification of a target sequence. Each cycle doubles the amount of target DNA. The reaction is specific due to primer design (primers only bind to complementary sequences in the template).

Step-by-Step Procedure

Part A: PCR Setup (15 minutes)

1. Label PCR tubes: Positive, Negative, Marker control
2. Prepare master mix (for 3 reactions):
 - 15 µL 10× buffer
 - 3 µL dNTP mix
 - 3 µL Forward primer
 - 3 µL Reverse primer
 - 0.75 µL Taq polymerase
 - 122.25 µL water
 - Total: 147 µL master mix
3. Aliquot 49 µL master mix into each tube
4. Add 1 µL template DNA to positive reaction tube
5. Add 1 µL water to negative control tube
6. Vortex briefly and spin down

Part B: PCR Cycling (90 minutes)

7. Place tubes in thermal cycler

Step	Temperature	Time	Purpose	Cycles
Initial denaturation	95°C	3 min	Complete denaturation	1
Denaturation	95°C	30 sec	Separate strands	30
Annealing	55°C	30 sec	Primers bind	30
Extension	72°C	1 min	Taq polymerase extends	30
Final extension	72°C	5 min	Complete synthesis	1
Hold	4°C	∞	Storage	—

Part C: Analysis by Gel Electrophoresis (30 minutes)

8. Prepare 1.5% agarose gel (see Lab 03)
9. Mix 5 µL PCR product + 1 µL loading dye
10. Load samples: Ladder, Positive, Negative
11. Run at 100 V for 25-30 minutes
12. Visualize under UV light
13. Photograph gel

Expected Observations

Lane	Expected Result
Ladder	Multiple bands at known sizes
Positive	Single sharp band at expected product size
Negative	No band (no contamination)
Positive control	Band at expected size

Results Table

Lane	Sample	Expected Size	Observed Size	Band Intensity	Interpretation
1	Ladder	Multiple			Reference
2	Positive				
3	Negative	None			Clean/no contamination
4					

Calculations

Amplification Efficiency

For 30 cycles with 100% efficiency:

- Final copies = Initial copies $\times 2^{30}$

- $= 10 \text{ ng} \times (10^9 / 660 \times 3.2 \times 10^6) \times 2^{30}$
- $\approx$ millions of copies

Annealing Temperature

$T_m = 2(A+T) + 4(G+C) \text{ }^\circ\text{C}$ (for primers <20 nt)

Example: Primer sequence: 5'-GCGATCGATCGATCGATCG-3'

- $A+T = 4$; $G+C = 14$
- $T_m = 2(4) + 4(14) = 8 + 56 = 64^\circ\text{C}$
- Annealing temperature: $T_m - 5^\circ\text{C} = 59^\circ\text{C}$

Interpretation

1. Why is a negative control important?
2. What would happen if you forgot to add Taq polymerase?
3. How does the annealing temperature affect specificity?
4. Compare the mechanism of PCR with cellular DNA replication.

Troubleshooting

Problem	Possible Cause	Solution
No band in positive	Insufficient template; primer design issue	Increase template; redesign primers
Band in negative control	Contamination	Use fresh reagents; UV-treat workspace
Multiple bands	Non-specific annealing	Increase annealing temperature
Smear	Over-cycling; non-specific	Reduce cycles; optimize annealing temp
Weak band	Low template; old Taq	Increase template; use fresh Taq
Band at wrong size	Primer dimers; non-specific	Check primer design; optimize conditions

Post-Lab Questions

1. A researcher gets no PCR product. List 5 possible reasons.
2. Why is 72°C the optimal extension temperature for Taq polymerase?
3. How does Pfu polymerase differ from Taq? When would you choose Pfu?
4. Calculate the theoretical yield after 25 cycles starting from 10 molecules.
5. Design a simple PCR experiment to detect a specific gene.

Viva Questions

1. What temperature is used for denaturation and why?
2. Why are two primers needed for PCR?
3. What determines the size of the PCR product?
4. How does the annealing temperature affect PCR specificity?

5. Why is Taq polymerase used instead of *E. coli* DNA polymerase?

Expected Answers

1. **Possible causes:** No template DNA, incorrect primers, missing Taq, insufficient cycles, inhibitors in sample, incorrect annealing temperature.
2. **72°C** is the optimal temperature for Taq polymerase activity (isolated from *Thermus aquaticus*, a thermophilic bacterium).
3. **Pfu has proofreading** (3'→5' exonuclease) but is slower; use Pfu when high fidelity is needed (cloning, sequencing).
4. $10 \times 2^{25} = 10 \times 33,554,432 \approx 3.36 \times 10^8$ copies.
5. **Design primers flanking the target gene**, choose annealing temperature based on T_m , run PCR, analyze by gel electrophoresis.

Instructor Notes

- This is a wet-lab exercise requiring a thermal cycler
- Use a simple, well-characterized target (e.g., human Alu element, λ phage gene)
- Prepare reagents in advance; keep Taq on ice
- Discuss PCR optimization (annealing temperature, Mg^{2+} concentration)
- Emphasize the distinction between PCR and cellular replication
- Connect to Module 03 (prokaryotic replication) and Module 04 (eukaryotic replication)

Related Theory

- [Module 03: Prokaryotic Replication](#)
- [Module 04: Eukaryotic Replication](#)
- [Module 09: Experimental Methods](#)
- [Module 11: Clinical Applications](#)

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Lab 05: Replication Data Analysis

Experiment Title

Computational Analysis of DNA Replication Data

Learning Objectives

1. Analyze simulated DNA fiber data
2. Interpret Okazaki fragment sequencing data
3. Perform bioinformatics analysis of replication origins
4. Analyze replication timing data
5. Connect computational analysis to experimental validation

Background Theory

DNA Fiber Analysis

- Label cells with nucleotide analogs (IdU, CldU)
- Stretch DNA fibers on glass slides
- Measure track lengths → fork speed
- Count tracks → origin usage
- Analyze patterns → fork dynamics

Repli-seq

- Sort cells by S-phase stage (FACS)
- Sequence replicating DNA
- Map early vs. late replicating regions

Origin Mapping

- Identify DnaA box sequences (E. coli)
- Use GC-skew analysis for origin prediction
- Analyze Okazaki fragment distribution for origin localization

Materials

Computational Tools

- Web browser (for online bioinformatics tools)
- Text editor

- Calculator
- Provided datasets (below)

Equipment

- Computer with internet access
- Calculator
- Printer (optional)

Safety Considerations

- This is a computational exercise — no wet-lab hazards
- Follow institutional guidelines for computer use

Experimental Principle

Bioinformatics analysis of replication data involves sequence analysis, statistical interpretation, and data visualization. Students will use simulated datasets that mirror real experimental data to develop analytical skills.

Exercise 1: DNA Fiber Analysis Interpretation (30 minutes)

Dataset 1: Fiber Track Lengths

A researcher treated cells with IdU (20 min) followed by CldU (20 min) and performed DNA fiber analysis. The following track lengths were measured:

Wild-type cells:

Fiber #	IdU track (μm)	CldU track (μm)	Pattern
1	12.5	13.0	Symmetric bidirectional
2	8.0	8.2	Symmetric bidirectional
3	11.2	11.5	Symmetric bidirectional
4	0	15.5	Asymmetric (one fork only)
5	10.8	11.0	Symmetric bidirectional
6	14.2	14.5	Symmetric bidirectional
7	9.5	9.8	Symmetric bidirectional
8	11.0	0	Asymmetric (one fork only)

Drug-treated cells (Replication inhibitor):

Fiber #	IdU track (μm)	CldU track (μm)	Pattern
1	12.0	4.0	Asymmetric (fork slowing)

Fiber #	IdU track (μm)	CldU track (μm)	Pattern
2	11.5	3.5	Asymmetric (fork slowing)
3	10.8	3.8	Asymmetric (fork slowing)
4	12.2	0	Fork collapse
5	11.0	2.5	Asymmetric (fork slowing)

Questions:

1. Calculate the average fork speed for wild-type cells:

- Assume $1\ \mu\text{m} \approx 2\ \text{kb}$ for stretched DNA
- Pulse time = 20 minutes
- Speed = (track length $\times$ 2 kb/ μm) / 20 min

2. Compare fork speed between wild-type and drug-treated cells:

- Is the drug affecting fork speed or fork stability?

3. What does the "asymmetric" pattern (Fiber #4, #8) tell you about replication?

4. Predict what a "fork collapse" event would look like on a fiber:

Exercise 2: Okazaki Fragment Analysis (20 minutes)

Dataset 2: Okazaki Fragment Size Distribution

A researcher analyzed Okazaki fragments from two cell lines using alkaline gel electrophoresis:

Cell Line A (Normal):

Fragment Size Range	Relative Abundance
50-100 nt	10%
100-150 nt	45%
150-200 nt	35%
200-300 nt	10%
>300 nt	0%

Cell Line B (Mutant):

Fragment Size Range	Relative Abundance
50-100 nt	5%
100-150 nt	15%
150-200 nt	25%
200-300 nt	35%
>300 nt	20%

Questions:

1. What is the average Okazaki fragment size for each cell line?
2. Cell Line B has larger fragments. Which protein might be mutated?
 - Consider: Primase, RNase H, FEN1, DNA Ligase I
3. How would you confirm your hypothesis about the mutation in Cell Line B?
4. Design an experiment to test whether the mutation affects primer processing:
 - What would you measure?
 - What controls would you include?

Exercise 3: Origin Prediction by GC-Skew (25 minutes)

Dataset 3: GC-Skew Data

A researcher calculated GC-skew for a 100 kb region of a circular bacterial chromosome:

Position (kb):	0	10	20	30	40	50	60	70	80	90	100
GC-skew:	+0.05	+0.08	+0.12	+0.15	+0.02	-0.05	-0.12	-0.18	-0.15	-0.08	+0.05

Questions:

1. Plot the GC-skew values against position:
2. At what position does the GC-skew change from positive to negative?
3. What does this transition indicate about origin and terminus locations?
4. How does GC-skew analysis compare to experimental origin mapping methods?

Exercise 4: Replication Timing Analysis (25 minutes)

Dataset 4: Repli-seq Data

Gene expression and replication timing data for Chromosome 1 (partial):

Gene	Expression Level	Replication Timing	Gene Size (kb)
Gene A	High	Early-S	2.5
Gene B	High	Early-S	1.8
Gene C	Low	Late-S	15.0
Gene D	High	Early-S	3.2
Gene E	Low	Late-S	22.0
Gene F	High	Early-S	4.1
Gene G	Low	Late-S	18.5
Gene H	High	Early-S	2.9

Gene	Expression Level	Replication Timing	Gene Size (kb)
Gene I	Low	Late-S	25.0
Gene J	High	Early-S	3.5

Questions:

1. Calculate the correlation between gene expression and replication timing.
 - Assign numerical values: Early-S = 1, Late-S = 0
2. Calculate the correlation between gene size and replication timing.
3. What is the biological significance of these correlations?
4. Predict the replication timing of a newly discovered gene that is: a. Highly expressed, 5 kb b. Lowly expressed, 30 kb

Exercise 5: Mutation Consequence Analysis (20 minutes)

Dataset 5: Replication Gene Mutations

Mutation	Organism	Protein Affected	Consequence
G671S	E. coli	Pol III ϵ subunit	?
R110A	Human	PCNA (PIP box)	?
S122E	Human	MSH6	?
N925S	Human	TERT	?
C228T	Human	TERT promoter	?
E49A	E. coli	DnaA	?

Questions:

1. For each mutation, predict the phenotype:
 - Would replication be affected?
 - What aspect of replication would be impaired?
2. How would you test these predictions experimentally?
3. Which of these mutations might be associated with cancer? Explain.

Exercise 6: Comparative Analysis (15 minutes)

Dataset 6: Replication Protein Comparison

Protein Feature	E. coli	Yeast	Human
Helicase	DnaB	MCM2-7	MCM2-7
Sliding clamp	β_2	PCNA	PCNA

Protein Feature	E. coli	Yeast	Human
Primase	DnaG	Pol α /primase	Pol α /primase
Leading Pol	Pol III	Pol ϵ	Pol ϵ
Lagging Pol	Pol III	Pol δ	Pol δ

Questions:

1. Which features are conserved? What does this suggest about evolution?
2. Why do eukaryotes use different polymerases for leading and lagging strands?
3. If you discovered a new organism with a circular genome but PCNA-like sliding clamp, what would you predict about its replication system?

Expected Results

Exercise 1 Answers (Fiber Analysis)

1. **Wild-type speed:** Average IdU track = 11.0 μm ; $11.0 \times 2 \text{ kb}/\mu\text{m} = 22 \text{ kb}$; $22 \text{ kb} / 20 \text{ min} = 1.1 \text{ kb/min} \approx 18 \text{ nt/s}$
2. **Drug effect:** CldU tracks shorter (4.0 vs 13.0 μm) $\rightarrow$ drug slows fork progression after IdU pulse
3. **Asymmetric pattern:** Only one fork visible $\rightarrow$ may be a dormant origin or one-directional fork
4. **Fork collapse:** Abrupt termination of one track $\rightarrow$ DSB at fork

Exercise 2 Answers (Okazaki Fragments)

1. **Normal:** Average $\sim 150 \text{ nt}$; **Mutant:** Average $\sim 250 \text{ nt}$
2. **Likely FEN1 or RNase H** — impaired primer processing $\rightarrow$ larger fragments
3. **Confirm by sequencing:** Look for RNA sequence remnants at fragment junctions

Exercise 3 Answers (GC-Skew)

1. **Transition at $\sim 45 \text{ kb}$** (positive $\rightarrow$ negative)
2. **Origin at $\sim 45 \text{ kb}$** (transition point); **Terminus at $\sim 95 \text{ kb}$** (opposite transition)
3. **GC-skew reflects strand-specific mutation patterns** caused by asymmetric replication

Exercise 4 Answers (Replication Timing)

1. **Positive correlation:** Highly expressed genes tend to replicate early
2. **Negative correlation:** Large genes tend to replicate late
3. **Biological significance:** Active chromatin is more accessible to replication machinery

Exercise 5 Answers (Mutations)

1. **G671S (ϵ):** Loss of proofreading $\rightarrow$ increased mutation rate
2. **R110A (PCNA):** Loss of PIP interaction $\rightarrow$ impaired polymerase switching
3. **S122E (MSH6):** Defective mismatch repair $\rightarrow$ microsatellite instability
4. **N925S (TERT):** Altered telomerase activity $\rightarrow$ telomere dysfunction

5. **C228T (TERT promoter)**: Created ETS binding site → telomerase reactivation → cancer

6. **E49A (DnaA)**: Defective ATP binding → impaired origin firing

Post-Lab Questions

1. Design a complete bioinformatics workflow to identify replication origins in a newly sequenced bacterial genome.
2. How would you use DNA fiber analysis to test whether a drug is a replication fork inhibitor?
3. Explain how replication timing analysis can inform our understanding of gene regulation.
4. Why is the distinction between "fork slowing" and "fork collapse" important in cancer biology?

Viva Questions

1. What information can DNA fiber analysis provide about replication?
2. How does Okazaki fragment size differ between prokaryotes and eukaryotes? Why?
3. What is the biological significance of replication timing?
4. How can bioinformatics predict replication origins without experimental data?
5. What are the limitations of computational replication analysis?

Instructor Notes

- This is a computational exercise — no wet-lab equipment needed
- Provide printed datasets or digital files
- Allow time for students to work through each exercise
- Discuss the connection between computational predictions and experimental validation
- Use this lab to introduce students to bioinformatics approaches
- Connect to Modules 09, 10, and 11

Related Theory

- [Module 09: Experimental Methods](#)
- [Module 10: Data Analysis](#)
- [Module 12: Advanced Topics](#)

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Lab 06: Capstone Exercise — Integrated Replication Analysis

Experiment Title

Integrated Analysis of DNA Replication: From Sequence to Clinical Application

Learning Objectives

1. Integrate knowledge from all previous modules
2. Analyze a complete replication scenario from initiation to termination
3. Connect molecular mechanisms to disease phenotypes
4. Design experiments to test replication-related hypotheses
5. Present findings in a scientific format

Background Theory

This capstone exercise brings together:

- Replication initiation (oriC/ORC)
- Elongation (replisome, polymerases)
- Primer processing (Okazaki fragments)
- Fidelity (proofreading, mismatch repair)
- Termination
- Telomere biology
- Replication stress and checkpoints
- Clinical applications

Materials

- Computer with internet access
- Provided case datasets (below)
- Whiteboard/paper for diagrams
- Calculator

Safety Considerations

- This is a computational/analytical exercise — no wet-lab hazards
- Follow institutional guidelines for computer and workspace use

Experimental Principle

Real-world replication problems require integrating multiple concepts. This exercise presents realistic scenarios that require students to apply knowledge from all modules to solve problems.

Part A: The *E. coli* Replication Scenario (30 minutes)

Scenario

A researcher isolates a new *E. coli* mutant (strain Δ dnaX) that has a frameshift mutation in the dnaX gene, which encodes the τ and γ subunits of the clamp loader complex.

Data Provided

Wild-type vs. Mutant Phenotype:

Parameter	Wild-type	Δ dnaX mutant
Growth rate (doubling time)	20 min	90 min
Mutation rate	10^{-9} /bp/div	10^{-7} /bp/div
Okazaki fragment size	1,500 nt	200 nt
Pol III processivity	>50,000 nt	~50 nt
Leading strand synthesis	Continuous	Continuous but slow
Lagging strand synthesis	Continuous cycling	Frequent polymerase dissociation

Questions

- Explain why the mutant has a slower growth rate.**
 - Consider the role of the clamp loader in processivity
- Explain why Okazaki fragments are shorter in the mutant.**
 - What happens when the sliding clamp cannot be loaded efficiently?
- Why is the mutation rate 100-fold higher in the mutant?**
 - How does processivity affect fidelity?
- Design an experiment to test whether adding back functional τ subunit rescues the phenotype.**
- Predict the phenotype if only the γ subunit (not τ) were absent.**

Part B: The Cancer Patient Scenario (30 minutes)

Scenario

A 45-year-old patient presents with:

- History of colorectal cancer (age 38)
- Endometrial cancer (age 42)

- Mother had ovarian cancer (age 50)
- Father had gastric cancer (age 55)
- Tumor testing shows microsatellite instability (MSI-High)

Data Provided

Tumor Molecular Profile:

Test	Result
MSI testing	MSI-High
MLH1 expression	Absent in tumor
MLH1 promoter methylation	Negative
MLH1 germline sequencing	Heterozygous frameshift mutation
MSH2 expression	Present
MSH6 expression	Present
PMS2 expression	Reduced (likely due to MLH1 loss)
BRAF V600E	Negative

Questions

1. **What is the most likely diagnosis?**
 - Explain the molecular basis
2. **Why is MLH1 promoter methylation negative?**
 - How does this help distinguish sporadic from hereditary cancer?
3. **Explain the mechanism by which MLH1 loss leads to MSI.**
 - What mismatch repair step is affected?
4. **Recommend genetic testing for the patient's siblings.**
 - What is the recurrence risk?
5. **What immunotherapy might be effective for this patient?**
 - Explain the mechanism
6. **Design a surveillance protocol for this patient.**

Part C: The Telomere Mystery (20 minutes)

Scenario

A researcher is studying two cell lines:

- **Cell line X:** Normal telomere length, finite lifespan (60 population doublings)
- **Cell line Y:** Very short telomeres (~2 kb), enters senescence after 15 population doublings

The researcher transfects Cell line Y with a retrovirus expressing TERT (telomerase catalytic subunit).

After TERT Expression:

Parameter	Cell line Y (before)	Cell line Y (after TERT)
Telomere length	2 kb	Stable at 10 kb after 20 passages
Population doublings	15	> 500 (still dividing)
p53 status	Wild type	Wild type
Senescence markers	Positive (p21, p16)	Negative
Karyotype	Normal	Normal initially, then abnormalities
Soft agar growth	Negative	Positive after 100 passages

Questions

1. Explain how TERT expression rescues the senescence phenotype.
2. Why does the karyotype eventually become abnormal despite TERT expression?
3. What does the soft agar growth result indicate?
 - Has the cell line become transformed?
4. What is the clinical significance of these findings for cancer therapy?
5. Design an experiment to test whether these cells are truly immortalized vs. transformed.

Part D: Replication Stress and Drug Response (20 minutes)

Scenario

A pharmaceutical company is developing a new drug (Drug Z) that inhibits a protein involved in replication fork restart. You are asked to analyze its mechanism.

Data Provided

Drug Z Effects:

Parameter	Control	Drug Z (low dose)	Drug Z (high dose)
Cell viability (24h)	100%	85%	30%
γH2AX foci	Low	Moderate	High
pRPA foci	Low	High	Very high
BRCA1 status	Wild type	Wild type	Wild type
Cell cycle arrest	—	S-phase	S-phase + G2/M
Fork speed (fiber assay)	50 nt/s	45 nt/s	30 nt/s
Fork reversal events	Low	High	Very high

Questions

1. Based on the data, what is the likely target of Drug Z?

- Which protein involved in fork restart?
- 2. **Why does Drug Z cause more toxicity in BRCA1-wild-type cells than BRCA1-deficient cells would?**
 - Consider the synthetic lethal interaction
- 3. **Would Drug Z be effective as a monotherapy? Why or why not?**
- 4. **Design a combination therapy strategy using Drug Z.**
- 5. **What biomarker would you use to select patients for Drug Z therapy?**

Expected Results

Part A Key Answers

1. **Growth rate:** Without functional clamp loader, Pol III processivity drops dramatically → each Okazaki fragment takes much longer to synthesize → overall replication is much slower
2. **Okazaki fragment size:** Without efficient clamp loading, Pol III dissociates after ~200 nt (vs. >50,000 nt with clamp)
3. **Mutation rate:** Lower processivity → more frequent polymerase cycling → more opportunities for misincorporation at primer junctions
4. **Rescue experiment:** Transform Δ dnaX with plasmid expressing wild-type τ → should restore normal growth rate and Okazaki fragment size
5. **γ -only mutant:** Less severe — τ -specific functions (dimerization, helicase coupling) would be affected, but basic clamp loading (γ function) would remain

Part B Key Answers

1. **Diagnosis:** Lynch syndrome (HNPCC) — germline MLH1 mutation with loss of heterozygosity in tumor
2. **MLH1 methylation negative:** Sporadic MSI-H typically shows MLH1 promoter methylation; negative methylation + germline mutation = hereditary
3. **MLH1 loss → MSI:** MLH1-PMS2 (MutL α) complex needed for strand-directed mismatch repair; loss → cannot repair replication errors at microsatellites
4. **Recurrence risk:** 50% for siblings (autosomal dominant); genetic testing recommended
5. **Immunotherapy:** Anti-PD-1 (pembrolizumab) — MSI-H tumors have high neoantigen load → respond to immune checkpoint inhibitors
6. **Surveillance:** Colonoscopy annually from age 20-25; endometrial screening; consider prophylactic surgery

Part C Key Answers

1. **TERT rescue:** TERT extends telomeres → prevents critically short telomeres → prevents senescence entry
2. **Karyotype abnormalities:** Telomerase prevents telomere-driven crisis but does not prevent other forms of genomic instability; cells can still acquire mutations during extended proliferation
3. **Soft agar growth:** Anchorage-independent growth = hallmark of transformation; cells have become cancerous

4. **Clinical significance:** Telomerase inhibition could prevent cancer cell immortalization; telomerase reactivation in normal cells could promote cancer
5. **Experiment:** Test for tumor formation in nude mice (tumorigenicity assay); karyotype analysis; p53 status

Part D Key Answers

1. **Target:** Likely a fork restart protein (e.g., RAD51, SMARCAL1, or BRCA2) — Drug Z increases fork reversal events
2. **BRCA1 consideration:** BRCA1-deficient cells already have fork restart defects → Drug Z may add little additional toxicity in BRCA1-deficient cells (but see data — actually, BRCA1 WT cells show more toxicity because Drug Z specifically targets fork restart)
3. **Monotherapy:** Likely insufficient — would need combination with DNA-damaging agents to exploit the fork restart defect
4. **Combination:** Drug Z + PARP inhibitor or platinum-based chemotherapy → exploit synthetic lethality
5. **Biomarker:** γ H2AX foci, pRPA foci, fork reversal events in tumor biopsies

Post-Lab Questions

1. How do the three scenarios connect to each other in terms of replication biology?
2. What common theme emerges across all four scenarios?
3. If you were a pharmaceutical researcher, which scenario would you focus on for drug development? Why?
4. How would you design a comprehensive replication profiling panel for a cancer center?

Viva Questions

1. Explain the connection between replication fidelity and cancer.
2. How does the clamp loader mutation affect both replication speed and accuracy?
3. Why does MLH1 loss specifically cause microsatellite instability?
4. How do telomere biology and replication stress interact in cancer development?
5. What is the future of replication-targeted cancer therapy?

Instructor Notes

- This is the culminating exercise — assess all learning objectives
- Allow students to work in groups or individually
- Present findings in a short oral presentation (5 minutes per group)
- Grade on scientific reasoning, not just correct answers
- Discuss how different concepts integrate
- Connect to all previous modules and labs

Related Theory

- [Module 03: Prokaryotic Replication](#)

- [Module 04: Eukaryotic Replication](#)
- [Module 05: Fidelity and Repair](#)
- [Module 06: Telomeres](#)
- [Module 07: Replication Stress](#)
- [Module 11: Clinical Applications](#)
- [Module 12: Advanced Topics](#)